Lob's theorem from the derivability conditions

The Open Logic Project contributors

openlogic:incompleteness/incompleteness-provability:local-snapshot-20261013; CC BY 4.0

Editorial note. The numbered labels L-1 to L-12 inside the proof span are internal equation labels of the counted span.

Editorial note. The derivability conditions P1 to P3 are cited from a companion section of the source; they are declared as a standard prerequisite rather than an in-unit reference.

Editorial scope. One original theorem of the unit, counted once: Lob's theorem for theories whose derivability predicate satisfies P1 to P3. The counted theorem and its author proof, including the labelled derivation chain, are retained; the informal Santa Claus heuristic dialogue before the proof is not counted.

Allowed standard prerequisites (editorial).

- An axiomatizable theory T extending Q with a derivability predicate satisfying the Hilbert-Bernays-Lob conditions P1 to P3: from a derivation of A one obtains a derivation of the provability of A; from a derivation of the provability of A-implies-B one obtains a derivation of the provability of A implies the provability of B; and from a derivation of A one obtains a derivation of the provability of the provability of A.
- The fixed-point lemma for Q, used to obtain a sentence D equivalent in T to the provability of D implies A.

Source reference. Original lemma fixed-point: sources/olp-incompleteness-incompleteness-provability/pinned-original-source.tex, source lines 616–616. The proof applies the fixed-point lemma to the formula provability-of-y implies A. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The lemma is proved in the same unit and is published as corpus card X2-incompleteness-incompleteness-provability-fixed-point.

Theorem 0.1. *Let* ThT *be axiomatizable theory extending* ThQ*, and suppose* Prov[ThT](y) *is a formula satisfying conditions P1–P3 from ?? . If* ThT *derives* Prov[ThT](gn A) $\to A$*, then in fact* ThT *derives* A*.*

Proof of 0.1 . Suppose A is sentence such that ThT derives Prov[ThT](gn A)$\to$ A. Let $B(y)$ be the formula Prov[ThT]$(y) \to A$, and use the fixed-point lemma to find sentence D such that ThT derives $D \leftrightarrow B(\text{gn}\, D)$. Then each of the

following is derivable in ThT:

$D \leftrightarrow (\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to A)$ (1)

D is a fixed point of $B(y)$

$D \to (\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to A)$ (2)

from 1

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D \to (\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to A))$ (3)

from 2 by condition P1

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to \mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, \mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to A)$ (4)

from 3 using condition P2

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to (\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, \mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D)) \to \mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, A))$ (5)

from 4 using P2 again

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to \mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, \mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D))$ (6)

by derivability condition P3

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to \mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, A)$ (7)

from 5 and 6

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, A) \to A$ (8)

by assumption of the theorem

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to A$ (9)

from 7 and 8

$(\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D) \to A) \to D$ (10)

from 1

D (11)

from 9 and 10

$\mathrm{Prov}[\mathrm{Th}T](\mathrm{gn}\, D)$ (12)

from 11 by condition P1

A from 8 and 12

□